



UNITED STATES PATENT AND TRADEMARK OFFICE

20120329739

UNITED STATES DEPARTMENT OF COMMERCE
United States Patent and Trademark Office
Address: COMMISSIONER FOR PATENTS
P.O. Box 1450
Alexandria, Virginia 22313-1450
www.uspto.gov

APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
13/582,110	08/31/2012	Hyoungh Jun Kim	K5675.0142/P112	5961

24998 7590 08/06/2014
DICKSTEIN SHAPIRO LLP
1825 EYE STREET NW
Washington, DC 20006-5403

EXAMINER

SHAO, YIH HORNG

ART UNIT PAPER NUMBER

1672

DATE MAILED: 08/06/2014

Determination of Patent Term Adjustment under 35 U.S.C. 154 (b)

(Applications filed on or after May 29, 2000)

The Office has discontinued providing a Patent Term Adjustment (PTA) calculation with the Notice of Allowance.

Section 1(h)(2) of the AIA Technical Corrections Act amended 35 U.S.C. 154(b)(3)(B)(i) to eliminate the requirement that the Office provide a patent term adjustment determination with the notice of allowance. See Revisions to Patent Term Adjustment, 78 Fed. Reg. 19416, 19417 (Apr. 1, 2013). Therefore, the Office is no longer providing an initial patent term adjustment determination with the notice of allowance. The Office will continue to provide a patent term adjustment determination with the Issue Notification Letter that is mailed to applicant approximately three weeks prior to the issue date of the patent, and will include the patent term adjustment on the patent. Any request for reconsideration of the patent term adjustment determination (or reinstatement of patent term adjustment) should follow the process outlined in 37 CFR 1.705.

Any questions regarding the Patent Term Extension or Adjustment determination should be directed to the Office of Patent Legal Administration at (571)-272-7702. Questions relating to issue and publication fee payments should be directed to the Customer Service Center of the Office of Patent Publication at 1-(888)-786-0101 or (571)-272-4200.

OMB Clearance and PRA Burden Statement for PTOL-85 Part B

The Paperwork Reduction Act (PRA) of 1995 requires Federal agencies to obtain Office of Management and Budget approval before requesting most types of information from the public. When OMB approves an agency request to collect information from the public, OMB (i) provides a valid OMB Control Number and expiration date for the agency to display on the instrument that will be used to collect the information and (ii) requires the agency to inform the public about the OMB Control Number's legal significance in accordance with 5 CFR 1320.5(b).

The information collected by PTOL-85 Part B is required by 37 CFR 1.311. The information is required to obtain or retain a benefit by the public which is to file (and by the USPTO to process) an application. Confidentiality is governed by 35 U.S.C. 122 and 37 CFR 1.14. This collection is estimated to take 12 minutes to complete, including gathering, preparing, and submitting the completed application form to the USPTO. Time will vary depending upon the individual case. Any comments on the amount of time you require to complete this form and/or suggestions for reducing this burden, should be sent to the Chief Information Officer, U.S. Patent and Trademark Office, U.S. Department of Commerce, P.O. Box 1450, Alexandria, Virginia 22313-1450. DO NOT SEND FEES OR COMPLETED FORMS TO THIS ADDRESS. SEND TO: Commissioner for Patents, P.O. Box 1450, Alexandria, Virginia 22313-1450. Under the Paperwork Reduction Act of 1995, no persons are required to respond to a collection of information unless it displays a valid OMB control number.

Privacy Act Statement

The Privacy Act of 1974 (P.L. 93-579) requires that you be given certain information in connection with your submission of the attached form related to a patent application or patent. Accordingly, pursuant to the requirements of the Act, please be advised that: (1) the general authority for the collection of this information is 35 U.S.C. 2(b)(2); (2) furnishing of the information solicited is voluntary; and (3) the principal purpose for which the information is used by the U.S. Patent and Trademark Office is to process and/or examine your submission related to a patent application or patent. If you do not furnish the requested information, the U.S. Patent and Trademark Office may not be able to process and/or examine your submission, which may result in termination of proceedings or abandonment of the application or expiration of the patent.

The information provided by you in this form will be subject to the following routine uses:

1. The information on this form will be treated confidentially to the extent allowed under the Freedom of Information Act (5 U.S.C. 552) and the Privacy Act (5 U.S.C. 552a). Records from this system of records may be disclosed to the Department of Justice to determine whether disclosure of these records is required by the Freedom of Information Act.
2. A record from this system of records may be disclosed, as a routine use, in the course of presenting evidence to a court, magistrate, or administrative tribunal, including disclosures to opposing counsel in the course of settlement negotiations.
3. A record in this system of records may be disclosed, as a routine use, to a Member of Congress submitting a request involving an individual, to whom the record pertains, when the individual has requested assistance from the Member with respect to the subject matter of the record.
4. A record in this system of records may be disclosed, as a routine use, to a contractor of the Agency having need for the information in order to perform a contract. Recipients of information shall be required to comply with the requirements of the Privacy Act of 1974, as amended, pursuant to 5 U.S.C. 552a(m).
5. A record related to an International Application filed under the Patent Cooperation Treaty in this system of records may be disclosed, as a routine use, to the International Bureau of the World Intellectual Property Organization, pursuant to the Patent Cooperation Treaty.
6. A record in this system of records may be disclosed, as a routine use, to another federal agency for purposes of National Security review (35 U.S.C. 181) and for review pursuant to the Atomic Energy Act (42 U.S.C. 218(c)).
7. A record from this system of records may be disclosed, as a routine use, to the Administrator, General Services, or his/her designee, during an inspection of records conducted by GSA as part of that agency's responsibility to recommend improvements in records management practices and programs, under authority of 44 U.S.C. 2904 and 2906. Such disclosure shall be made in accordance with the GSA regulations governing inspection of records for this purpose, and any other relevant (i.e., GSA or Commerce) directive. Such disclosure shall not be used to make determinations about individuals.
8. A record from this system of records may be disclosed, as a routine use, to the public after either publication of the application pursuant to 35 U.S.C. 122(b) or issuance of a patent pursuant to 35 U.S.C. 151. Further, a record may be disclosed, subject to the limitations of 37 CFR 1.14, as a routine use, to the public if the record was filed in an application which became abandoned or in which the proceedings were terminated and which application is referenced by either a published application, an application open to public inspection or an issued patent.
9. A record from this system of records may be disclosed, as a routine use, to a Federal, State, or local law enforcement agency, if the USPTO becomes aware of a violation or potential violation of law or regulation.

Examiner-Initiated Interview Summary	Application No.	Applicant(s)	
	13/582,110	KIM ET AL.	
	Examiner	Art Unit	
	YIH-HORNG SHIAO	1672	

All participants (applicant, applicant's representative, PTO personnel):

(1) YIH-HORNG SHIAO (3) _____

(2) Jonathan England (4) _____

Date of Interview: 31 July 2014.

Type: ☒ Telephonic ☐ Video Conference
☐ Personal [copy given to: ☐ applicant ☐ applicant's representative]

Exhibit shown or demonstration conducted: ☐ Yes ☒ No.
If Yes, brief description: _____.

Issues Discussed ☐ 101 ☒ 112 ☐ 102 ☒ 103 ☐ Others
(For each of the checked box(es) above, please describe below the issue and detailed description of the discussion)

Claim(s) discussed: 11.

Identification of prior art discussed: _____.

Substance of Interview
(For each issue discussed, provide a detailed description and indicate if agreement was reached. Some topics may include: identification or clarification of a reference or a portion thereof, claim interpretation, proposed amendments, arguments of any applied references etc...)

Applicants agreed with Examiner-proposed amendments to overcome potential 112(b) and 103(a) rejections and the claims would then be allowable..

Applicant recordation instructions: It is not necessary for applicant to provide a separate record of the substance of interview.

Examiner recordation instructions: Examiners must summarize the substance of any interview of record. A complete and proper recordation of the substance of an interview should include the items listed in MPEP 713.04 for complete and proper recordation including the identification of the general thrust of each argument or issue discussed, a general indication of any other pertinent matters discussed regarding patentability and the general results or outcome of the interview, to include an indication as to whether or not agreement was reached on the issues raised.

☐ Attachment

YIH-HORNG SHIAO/ Examiner, Art Unit 1672	
---	--

Notice of Allowability

Application No.
13/582,110

Examiner
YIH-HORNG SHIAO

Applicant(s)
KIM ET AL.

Art Unit
1672

AIA (First Inventor to
File) Status
No

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address--

All claims being allowable, PROSECUTION ON THE MERITS IS (OR REMAINS) CLOSED in this application. If not included herewith (or previously mailed), a Notice of Allowance (PTOL-85) or other appropriate communication will be mailed in due course. **THIS NOTICE OF ALLOWABILITY IS NOT A GRANT OF PATENT RIGHTS.** This application is subject to withdrawal from issue at the initiative of the Office or upon petition by the applicant. See 37 CFR 1.313 and MPEP 1308.

1. ☐ This communication is responsive to 06-19-2014.
☐ A declaration(s)/affidavit(s) under 37 CFR 1.130(b) was/were filed on _____.
2. ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.
3. ☐ The allowed claim(s) is/are 11 and 13-18. As a result of the allowed claim(s), you may be eligible to benefit from the Patent Prosecution Highway program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to PPHfeedback@uspto.gov.
4. ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

- a) ☐ All b) ☐ Some *c) ☐ None of the:
1. ☐ Certified copies of the priority documents have been received.
 2. ☐ Certified copies of the priority documents have been received in Application No. _____.
 3. ☐ Copies of the certified copies of the priority documents have been received in this national stage application from the International Bureau (PCT Rule 17.2(a)).

* Certified copies not received: _____.

Applicant has THREE MONTHS FROM THE "MAILING DATE" of this communication to file a reply complying with the requirements noted below. Failure to timely comply will result in ABANDONMENT of this application.

THIS THREE-MONTH PERIOD IS NOT EXTENDABLE.

5. ☐ CORRECTED DRAWINGS (as "replacement sheets") must be submitted.
☐ including changes required by the attached Examiner's Amendment / Comment or in the Office action of Paper No./Mail Date _____.
Identifying indicia such as the application number (see 37 CFR 1.84(c)) should be written on the drawings in the front (not the back) of each sheet. Replacement sheet(s) should be labeled as such in the header according to 37 CFR 1.121(d).
6. ☐ DEPOSIT OF and/or INFORMATION about the deposit of BIOLOGICAL MATERIAL must be submitted. Note the attached Examiner's comment regarding REQUIREMENT FOR THE DEPOSIT OF BIOLOGICAL MATERIAL.

Attachment(s)

1. ☐ Notice of References Cited (PTO-892)
2. ☒ Information Disclosure Statements (PTO/SB/08),
Paper No./Mail Date 04-03-2014
3. ☐ Examiner's Comment Regarding Requirement for Deposit of Biological Material
4. ☒ Interview Summary (PTO-413),
Paper No./Mail Date 07-31-2014
5. ☒ Examiner's Amendment/Comment
6. ☒ Examiner's Statement of Reasons for Allowance
7. ☐ Other _____

/YIH-HORNG SHIAO/
Examiner, Art Unit 1672

/BRANDON FETTEROLF/
Supervisory Patent Examiner, Art Unit 1672

DETAILED ACTION

The present application is being examined under the pre-AIA first to invent provisions. Applicants' amendment filed on June 19, 2014 has been entered. Claims 1-10 are cancelled. Claim 12 is cancelled after Examiner's amendment. Claims 14-18 are newly amended. Claims 11 and 13-18 are currently under examination and allowed in this Office Action.

Priority

This application is a 371 of PCT/KR2010/001360 filed on March 04, 2010.

Information Disclosure Statement

The information disclosure statement (IDS) filed on April 03, 2014 has been considered.

Withdrawn Rejections

The objection of claim 7 because of improper format of transitional phrase, as set forth on page 2 of the Non-Final Rejection mailed on March 20, 2014, is *withdrawn* in view of cancelled claim.

The objection of claims 9 and 12 because of improper format for dependent claim, as set forth on pages 2 and 3 of the Non-Final Rejection mailed on March 20, 2014, is *withdrawn* in view of cancelled claims 9 and 12.

The objection of claim 11 because of improper format for independent claim, as set forth on page 3 of the Non-Final Rejection mailed on March 20, 2014, is *withdrawn* in view of amended claim 11.

The rejection of claims 7-13 under 35 U.S.C. 112(a) or 35 U.S.C. 112 (pre-AIA), first paragraph, as set forth on pages 3-8 of the Non-Final Rejection mailed on March 20, 2014, is *withdrawn* in view of cancelled claims 7-10 and 12, and amended claims 11 and 13.

The rejection of claims 7-11 and 13 under pre-AIA 35 U.S.C. 102(b) as being anticipated by D'Amelio et al., as set forth on pages 8 and 9 of the Non-Final Rejection mailed on March 20, 2014, is *withdrawn* in view of cancelled claims 7-10 and amended claims 11 and 13.

The rejection of claims 7-13 under pre-AIA 35 U.S.C. 102(b) as being anticipated by Cherdshewasart et al., as set forth on pages 9-11 of the Non-Final Rejection mailed on March 20, 2014, is *withdrawn* in view of cancelled claims 7-10 and 12, and amended claims 11 and 13.

The rejection of claims 11-13 under pre-AIA 35 U.S.C. 102(a) as being anticipated by Lee et al., as set forth on page 11 of the Non-Final Rejection mailed on March 20, 2014, is *withdrawn* in view of cancelled claim 12 and amended claims 11 and 13.

The rejection of claims 7-13 under pre-AIA 35 U.S.C. 103(a) as being unpatentable over D'Amelio et al., as set forth on pages 12-14 of the Non-Final

Rejection mailed on March 20, 2014, is *withdrawn* in view of cancelled claims 7-10 and 12, and amended claims 11 and 13.

The rejection of claims 7-10 under pre-AIA 35 U.S.C. 103(a) as being unpatentable over Sahastrayoga in view of Keung et al., as set forth on pages 14-16 of the Non-Final Rejection mailed on March 20, 2014, is *withdrawn* in view of cancelled claims 7-10.

Examiner's Amendment

An examiner's amendment to the record appears below. Should the changes and/or additions be unacceptable to applicant, an amendment may be filed as provided by 37 CFR 1.312. To ensure consideration of such an amendment, it MUST be submitted no later than the payment of the issue fee.

Authorization for this examiner's amendment was given in a telephone interview with Jonathan England on July 31, 2014

The amended claims are listed below.

11. A method for treating hair graying in a subject in need thereof, comprising:
administering to the subject an effective amount of a composition comprising 0.1-10% by weight of a puerarin based on a total weight of the composition to treat the subject's hair graying.

15. The method according to claim 11, ~~wherein the puerarin is administered in a form of a composition~~, wherein the composition comprises the puerarin in an amount of 0.1-5 wt% based on the total weight of the composition.

Art Unit: 1672

16. The method according to claim 11, ~~wherein the puerarin is administered in an form of a composition~~, wherein the composition comprises the puerarin in an amount of 0.1-2.5 wt% based on the total weight of the composition.

Allowable Subject Matter

The amended claim 11 is allowed. Claims 13-18, depending from claim 11, are also allowed.

The following is an examiner's statement of reasons for allowance: The amended claim 11, directed to a method for treating hair graying in a subject in need thereof, comprising: administering to the subject an effective amount of a composition comprising 0.1-10% by weight of a puerarin based on a total weight of the composition to treat the subject's hair graying, is free of prior art rejection. The closest prior art is cited in the Non-Final Rejection mailed on March 20, 2014, in which D'Amelio et al. (US Patent Application Publication 2005/0196475) disclosed a process of reducing wrinkles of the skin of a subject in need thereof comprising topically applying a composition comprising puerarin in about 0.03 wt% (from 1-5 wt% of powdered extract of *Pueraria candollei* var. *mirifica* A Shaw. & Suvat.) to the skin of the subject to reduce wrinkles of the skin; Cherdshewasart et al. (US Patent Application Publication 2004/0105900) disclosed a process of treating pre-menopausal women comprising administering a composition comprising about 0.004 wt% (from 4 wt% *Pueraria mirifica* extract and extract contains about 0.1 wt% puerarin) in a capsule thereby resulting in improved

Art Unit: 1672

smooth skin and healthy hair (or hair complexity) in the treated women compared with placebo group; Sahastrayoga (Central Council for Research in Ayurveda & Siddha, New Delhi, 1990) disclosed a composition comprising about 6% *Pueraria tuberosa* powder but the amount of puerarin in tuberosa is unknown; and Keung et al. (Phytochemistry 47:499-506, 1998) disclosed that *Pueraria lobata* contains about 8 wt% puerarin but there is no art teaching or suggesting treatment of hair graying and it is not in a composition. The reference did not explicitly disclose the limitation "a composition comprising 0.1-10% by weight of a puerarin based on a total weight of the composition", required by this instant application. In addition, Lee et al. (KR1020090066824, published on June 24, 2009) disclosed a process for treating a melanocyte comprising administering a composition comprising 1000 µg/ml (= about 0.1 wt%) puerarin to the melanocyte. Applicants' Arguments/Remarks (page 6, 3rd paragraph, filed on June 19, 2014) have been fully considered. Applicants argue "the composition of Lee inhibits the activity of the tyrosinase and therefore inhibits the production of melanin to cause skin whitening...Lee teaches away from hair darkening". *In response*, these arguments are found persuasive.

Any comments considered necessary by applicant must be submitted no later than the payment of the issue fee and, to avoid processing delays, should preferably accompany the issue fee. Such submissions should be clearly labeled "Comments on Statement of Reasons for Allowance."

Conclusion

Claims 11 and 13-18 are allowed.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to YIH-HORNG SHIAO whose telephone number is (571)272-7135. The examiner can normally be reached on Mon-Thur, 08:30 am to 07:00 pm EST.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Brandon Fetterolf can be reached on 571-272-2919. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/YIH-HORNG SHIAO/
Examiner, Art Unit 1672

/BRANDON FETTEROLF/
Supervisory Patent Examiner, Art Unit 1672